

Mastery, not multiple choice

An AI assessment pipeline for the EWSDEP 8670 prerequisite course — from coursebook to graded conversation inside Moodle

01

Rubric agent

02

Question agent

03

Moodle plugin

Team Nand2Tetris2 · MCU–NPS AI Learning Initiatives · September 2026

A quiz does not provide meaningful feedback.

13

lessons, each ending in a multiple-choice quiz

2–3 min

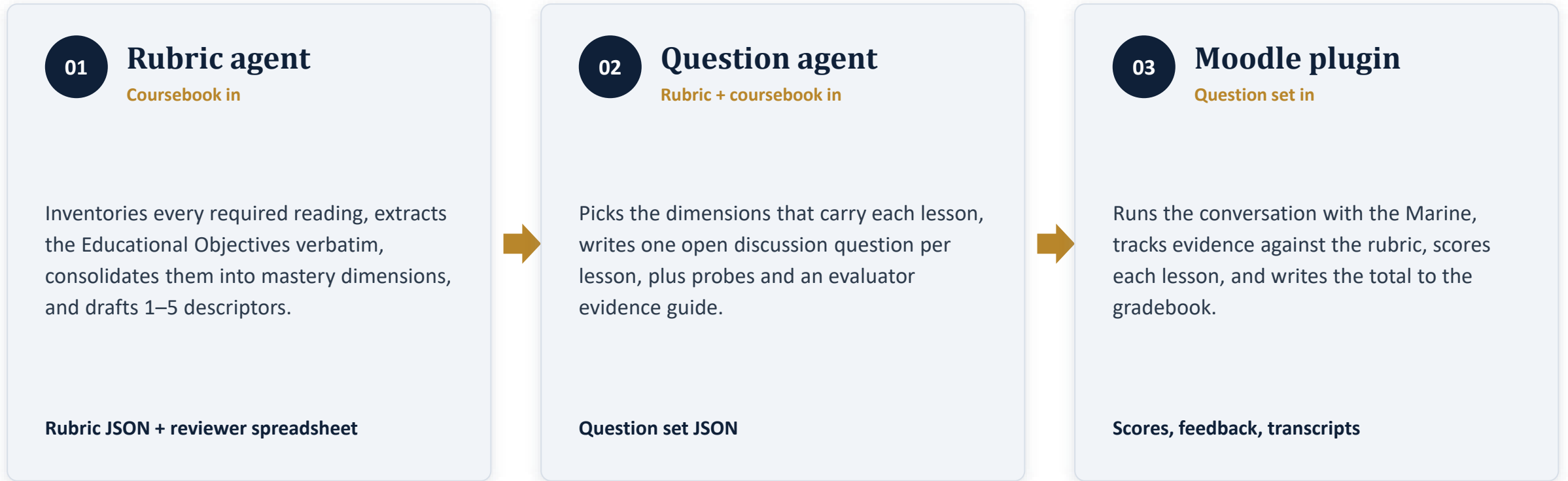
of written reasoning is all it takes to tell the difference

Multiple choice quizzes give you limited evidence about how a Marine thinks

From the use case

“Expand upon the limited utility of the quizzes to gauge student knowledge ... through a discussion with an AI agent, grounded in a rubric derived from the Educational Objectives ... coach and guide students to reach acceptable levels of knowledge mastery.”

Three stages, two files, one activity



Each stage hands off a plain JSON file. Swap any stage for a different model, vendor or team and the next one still works.

Agent 1 builds the rubric from the coursebook alone

1

Inventory before anything else

Every required reading per lesson: title, issuing organization, edition, URL, and whether it was actually retrieved.

2

Objectives extracted verbatim

Each Educational Objective gets a stable ID (L01-EO01) and a normalised reading of what it expects.

3

Consolidated into mastery dimensions

Related objectives group into dimensions (L01-MD01) with essential evidence and critical misconceptions named.

4

1–5 descriptors, observable and question-agnostic

Written so they can grade any constructed response for that lesson, not one specific prompt.

The rule that makes it trustworthy

It is not allowed to fill a gap from general knowledge.

If a required reading is behind a login, missing, or the wrong edition, it is recorded as a gap and the affected rubric content is marked provisional.

That is how the L03 problem surfaced: the coursebook assigns the 2022 National Defense Strategy, but a 2026 edition exists. The agent flagged the conflict instead of quietly picking one.

Agent 2 turns each lesson into one discussion question

1

Picks the dimensions that carry the lesson

The ones integrating several objectives and separating real understanding from recall — not the easiest to test.

2

One open prompt, 2–3 minutes

Explanation, comparison, diagnosis or a justified decision. No multiple choice, no definitions, no trivia.

3

Probes and an evidence guide, not an answer key

Strong evidence, partial evidence, misconceptions, and the conditions under which a response cannot be scored at all.

4

Refuses rather than fabricates

If the verified corpus cannot support a fair question, it returns a gap record instead of inventing one.

LESSON 1 — MCDP 7, LEARNING

“A unit treats learning as attendance at formal classes plus repetition of already familiar drills. Explain why that approach is incomplete for warfighting, and how individual Marines, instructors, and leaders should combine education, training, experience, and feedback to build continuous learning and adaptive competence.”

Targets L01-MD01 and L01-MD03 · built from L01-EO01, EO02, EO04

Every question carries its source citations, the objectives it targets, and the validation notes from the rubric it came from.

The handoff is one file — and that is the whole integration

question_text

the prompt the Marine sees

target_mastery_dimensions

what gets judged, and reported

follow_up_probes

what to ask when evidence is thin

evaluator_evidence_guide

strong · partial · misconceptions · insufficient

source_evidence

what the judgement traces back to

validation_notes

the caveats a reviewer must see

Anything that emits this file works

The plugin does not know or care what produced the JSON. The two agents run in ChatGPT Enterprise today because that is what the hackathon issued. They could run on a government-hosted model, in Azure, or on-premises, and nothing downstream changes.

Shipped with the contract

A written spec for the generating agent, and a validator that rejects a malformed set before anyone uploads it — duplicate IDs, thin rubrics, or a question that gives away its own answer.

Two ways into Moodle, from the same file

SIMPLE

An off-the-shelf question type

- `qtype_aitext` from the Moodle plugins directory, plus its two grading behaviours
- The same question set, converted to Moodle XML: 13 questions, each carrying its rubric as the grading prompt
- Graded through Moodle's core AI subsystem, straight into the gradebook
- One answer, one score. It cannot hold a conversation.

Zero lines of code. Stood up in an afternoon.

CUSTOM

A purpose-built activity

- `mod_masteryagent` — an activity module written for this use case
- Multi-turn: probes, challenges misconceptions, moves lesson to lesson on its own
- Keeps an evidence ledger against the rubric; scores each lesson and the run as a whole
- Instructor report pairs transcript and rubric for SME validation

No third-party dependencies.

The Marine gets a conversation, not a form

Opens with the lesson's question

Straight from the JSON, unchanged.

Probes the largest gap

One question at a time, chosen against the evidence the Marine has actually given.

Challenges misconceptions directly

Here: treating education and training as the same activity.

Bounded

Six replies per lesson, and it ends sooner once the evidence is there.

Lesson 1 – MCDP 7 Learning

View attempts

EVALUATOR

A unit treats learning as attendance at formal classes plus repetition of already familiar drills. Explain why that approach is incomplete for warfighting and how individual Marines, instructors, and leaders should combine education, training, experience, and feedback to build continuous learning and adaptive competence.

YOU

Learning is something that keeps going after the course ends. You pick things up from doing the job and from the schools you get sent to, and both of those matter. If a unit just runs the same drills over and over they will get faster at those drills but they are not really learning anything new.

EVALUATOR

You have the first piece: learning is development over time rather than seat time. But you are still treating education and training as the same activity, and the lesson turns on the difference. What does classroom study give a Marine that repetition on the range cannot?

Replies remaining: 5

Write your answer here

Send reply

End and score now

Scored per lesson, against the dimensions it came from

A score per lesson, and a total to the gradebook

Four points per lesson, three to clear the mastery threshold. The gradebook column scales with the number of lessons.

Feedback tied to named dimensions

Every judgement is reported against the mastery dimension it belongs to, so it traces back to an educational objective.

One judgement across the whole run

After the last lesson the agent looks at the body of work and names the pattern — here, the Marine can define concepts but not show mechanism.

Every grade flagged AI-provisional

It reaches the gradebook, and it is marked as awaiting subject matter expert validation.

Score: 9 of 12

Lessons at or above the mastery threshold: 2 of 3.

AI-provisional assessment, pending subject matter expert validation.

Across these three lessons your strongest work is conceptual: you explain what a framework is for and you qualify your sources carefully, which is rarer than it sounds. The pattern in what is missing is mechanism. You can say what integrated deterrence or continuous learning is, but you stop short of showing how it produces an effect, and in Lesson 2 that gap became an error about who commands. Before the next block, practise tracing one idea end to end: from national interest through strategy to the unit that acts on it, naming who decides at each step.

L01 MCDP 7 – Learning — 4/4

You established learning as continuous development and separated education from training clearly, then tied both to judgement under uncertainty. You also named what leaders owe their Marines rather than leaving development to the schoolhouse. The weakest part was feedback: you mentioned it but never connected it to changed performance.

Dimension	Judgement	Comment
L01-MD01	Met	Learning framed as continuous development for judgement under uncertainty.
L01-MD03	Met	Differentiated learner, instructor and leader responsibility.

Next step: Re-read MCDP 7 chapter 3 and come back with how feedback changes what a unit does next.

L02 Fundamentals of National Defense — 2/4

You can name the actors, but the chain from national interest to military action stayed vague. You placed the Chairman in the operational chain of command, which is the misconception this lesson is built to catch — he advises, he does not command. Rebuild the sequence from interest to strategy to unified action before you go further.

Dimension	Judgement	Comment
L02-MD01	Partial	Named the actors without tracing interest to strategy.
L02-MD05	Not met	Placed the Chairman in the operational chain of command.

Next step: Review the roles of the Chairman versus the combatant commander before the next lesson.

An instructor can audit every judgement

Rubric behind this assessment

L01 MCDP 7 – Learning

Assessment question A unit treats learning as attendance at formal classes plus repetition of already familiar drills. Explain why that approach is incomplete for warfighting and how individual Marines, instructors, and leaders should combine education, training, experience, and feedback to build continuous learning and adaptive competence.

Strong evidence

- Explains learning as continuous development of knowledge, skills, and attitudes for judgment and adaptation under uncertainty.
- Distinguishes education as study and intellectual development from training as learning by doing, while showing that both contribute to competence.
- Differentiates learner ownership, instructor facilitation and mentoring, and leader responsibility to set conditions, tailor experiences, model learning, and develop subordinates.
- Connects feedback and experience to changed performance rather than treating course completion as the outcome.

Partial evidence

- Correctly values both education and training but gives little explanation of how they interact.
- Names some role responsibilities but assigns most responsibility to one actor or does not connect them to adaptive warfighting.

Misconceptions and red flags

- Equates learning with memorization, course attendance, or repetition alone.
- Treats education and training as interchangeable or presents either as sufficient by itself.
- Assigns responsibility solely to instructors or institutions and omits leader example or learner ownership.

Insufficient evidence conditions

- The response is nonresponsive, too fragmentary, or too ambiguous to distinguish rubric levels.
- The learner gives labels or lists without enough explanation to demonstrate the targeted reasoning.
- A fair judgment would require a missing, inaccessible, or unresolved source that this prompt was designed not to require.

Validation notes from the question set

The transcript, the ledger, and the rubric side by side

An instructor sees what was said, which evidence the agent recorded, and the criteria it was working from.

Source citations travel with the rubric

Including the verification status — embedded, verified, version mismatch, or inaccessible.

The validation notes are shown, not buried

Including the open question about which National Defense Strategy edition AY27 should assess.

Nothing claims to be validated

The question set is marked DRAFT_PENDING_HUMAN_VALIDATION end to end, and the plugin surfaces that on every grade.

Against the use case requirements

Assess each lesson and the course as a whole	Built	Feedback on knowledge level against the rubric	Built
Embedded in the LMS, seamless to the student	Built	Coach and guide toward mastery	Partial
Easily updatable to current lessons and objectives	Built	Record a score, passed to the LMS	Built
Develop a rubric from the Educational Objectives	Built	Accessible from MCEN, .edu and personal devices	By design
Human validation of that rubric	Not yet		

“By design” means the plugin runs server-side, so it reaches any device that reaches Moodle — but it has not been tested on MCEN.

What is tested, and what is not

Tested

- Both integrations installed on a real Moodle: the off-the-shelf question type on the test LMS, the custom activity on Moodle 5.1.7
- 104 automated checks across the conversation engine, scoring, gradebook and failure paths
- Every page exercised through a real browser session
- Upgrade path verified from the first release to the current one
- Provider failures degrade gracefully — the Marine's answer is never lost

Not yet

- Backup and restore — declared unsupported rather than silently breaking course backups
- Moodle mobile app support
- Model calls are synchronous; no queue, no retry
- No pilot data — no Marine has used this yet
- The rubric itself remains an unvalidated draft

Questions?